

SIDNEY-RICHLAND, MT, USA

WMO: 727687

Lat: 47.717N

Lon: 104.183W

Elev: 1980

StdP: 13.67

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94028

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-12.9	-26.7	1.3	-13.0	-23.1	1.6	-9.6	29.6	27.9	26.7	27.4	5.4	190	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	94.1	67.9	90.6	67.2	87.7	66.5	72.0	87.3	69.9	85.2	68.0	83.0	9.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	105.0	80.2	64.3	97.0	77.5	63.0	92.6	75.7	37.0	87.8	35.1	85.0	33.3	82.8	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	23.6	20.3	-26.1	100.3	6.1	3.6	-30.5	102.9	-34.0	105.0	-37.5	107.0	-41.9	109.7
			-24.2	76.0	4.5	2.7	-27.5	77.9	-30.2	79.5	-32.7	81.0	-36.0	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.6	15.6	18.7	30.7	43.9	54.3	64.2	71.7	69.4	59.1	44.7	30.1	19.6	(6)
	DBStd	22.26	15.10	14.56	13.74	9.54	8.12	6.81	5.92	6.49	8.41	9.27	12.09	13.52	(7)
	HDD50	4594	1066	877	602	226	49	2	0	0	18	212	599	943	(8)
	HDD65	8304	1531	1297	1062	635	341	94	12	32	213	631	1048	1408	(9)
	CDD50	2268	0	0	5	42	182	426	671	601	292	47	2	0	(10)
	CDD65	504	0	0	0	1	10	69	218	168	37	1	0	0	(11)
	CDH74	7050	0	0	4	51	251	996	2754	2243	695	56	0	0	(12)
	CDH80	3011	0	0	0	12	61	342	1279	1029	277	11	0	0	(13)
	WSAvg	8.9	9.1	8.9	9.4	10.1	9.9	8.9	7.6	7.7	8.1	9.2	8.8	8.9	(14)
	PrecAvg	14.4	0.4	0.3	0.6	1.2	2.2	2.9	2.4	1.3	1.1	1.0	0.5	0.4	(15)
(16) Precipitation	PrecMax	20.6	1.3	1.4	2.2	3.8	5.4	5.7	8.1	4.7	5.7	3.9	2.6	1.1	(16)
	PrecMin	7.4	0.0	0.0	0.0	0.0	0.4	1.3	0.4	0.1	0.0	0.2	0.0	0.0	(17)
	PrecStd	2.9	0.3	0.3	0.4	1.0	1.3	1.3	1.7	0.9	1.1	0.8	0.6	0.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	54.8	71.9	81.3	86.2	94.2	99.8	99.1	93.4	81.1	64.2	50.2	(19)
		MCWB	40.6	42.8	51.8	55.8	61.2	68.1	69.6	68.4	65.2	58.0	48.9	40.2	(20)
	2%	DB	43.3	46.5	62.9	72.5	81.0	88.1	94.9	93.2	88.0	72.3	56.9	45.0	(21)
		MCWB	36.9	38.5	47.2	52.8	59.2	66.8	69.3	66.6	64.1	54.7	44.4	37.3	(22)
	5%	DB	39.0	41.5	55.2	66.4	75.4	83.6	90.9	90.1	81.9	65.7	52.0	40.2	(23)
		MCWB	34.4	35.6	43.4	49.7	57.1	64.6	68.6	66.3	61.1	51.2	42.2	34.5	(24)
	10%	DB	36.2	37.1	49.7	62.7	71.9	80.5	87.9	85.9	76.6	60.9	46.0	36.7	(25)
		MCWB	32.7	32.9	40.8	47.6	55.4	62.8	67.9	65.1	58.9	48.6	38.6	32.4	(26)
	0.4%	WB	41.2	44.0	52.0	57.1	65.5	72.3	76.3	73.4	68.7	60.3	50.1	41.1	(27)
		MCDB	48.5	53.2	69.7	76.1	77.7	86.9	90.7	88.6	86.0	76.0	62.8	48.6	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	39.3	47.8	53.5	61.6	69.3	73.3	70.3	65.7	55.8	45.7	37.5	(29)
		MCDB	43.0	46.5	61.1	70.4	76.5	83.1	88.0	87.0	82.8	69.8	55.9	44.1	(30)
	5%	WB	34.8	36.1	44.2	50.9	58.9	66.7	71.1	68.3	62.9	52.4	42.0	35.1	(31)
		MCDB	39.2	41.3	55.1	65.3	72.7	79.9	86.2	84.6	78.7	64.1	50.6	40.1	(32)
	10%	WB	32.6	33.3	40.9	48.3	56.5	64.2	69.0	66.7	60.3	49.6	39.2	32.6	(33)
		MCDB	36.1	37.2	48.8	60.8	69.1	77.2	83.8	82.2	74.6	59.8	45.6	36.8	(34)
(35) Mean Daily Temperature Range		MDBR	19.1	20.1	22.1	24.6	24.7	24.4	26.9	28.3	27.8	23.5	20.9	18.9	(35)
	5% DB	MCDBR	20.5	24.2	32.6	34.7	32.3	31.9	33.1	34.8	37.6	34.6	29.4	22.2	(36)
		MCWBR	15.8	17.2	20.0	18.7	16.4	15.5	14.5	14.9	18.2	19.4	19.1	16.2	(37)
	5% WB	MCDBR	20.2	23.1	31.8	31.8	28.6	28.4	28.8	30.1	33.3	31.7	27.6	21.8	(38)
		MCWBR	15.8	16.7	19.9	17.8	15.4	15.0	14.1	14.0	16.6	18.4	18.3	16.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.251	0.268	0.278	0.320	0.338	0.354	0.362	0.367	0.325	0.293	0.263	0.248	(40)
	taud		2.488	2.486	2.542	2.425	2.436	2.430	2.432	2.389	2.480	2.559	2.534	2.506	(41)
	Ebn at Noon		272	290	304	296	293	288	284	278	281	277	266	261	(42)
	Edh at Noon		19	24	27	34	35	35	35	35	29	23	18	17	(43)
(44) All-Sky Solar Radiation	RadAvg		486	802	1191	1557	1854	2066	2210	1811	1333	843	517	365	(44)
	RadStd		41	69	88	89	159	135	90	116	110	81	59	20	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page